CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mean of raw and tabulated data

1. Assertion (A): The mean of the first 9 natural numbers is 4.5. Reason (R): Mean $=$ $\frac{\text{sum of all the observations}}{\text{total number of observations}}$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A ICSE 2025, 1 mark

SOLUTION

$1+2+\ldots+9=45$, so the mean is $\frac{45}{9}=5$, not 4.5. A is false.

R is the correct formula for the mean.

Answer: (b)

2. For the data below, find (a) the mean, to the nearest whole number (b) the median. ICSE 2025, 3 marks

x	10	11	12	13	14	15	16
f	3	2	2	6	3	5	3

SOLUTION

(a) $\sum f=24$ and $\sum fx=30+22+24+78+42+75+48=319$. Mean $=\frac{319}{24}\approx 13.29$, which is 13 to the nearest whole number.

(b) Cumulative frequencies: 3, 5, 7, 13, 16, 21, 24. With $n=24$ (even), the median is the mean of the 12th and 13th values.

Values 8 to 13 are all 13, so the 12th and 13th values are both 13.

Answer: (a) mean $=13$ (b) median $=13$

3. The mean of the data below is 16. Find f. ICSE 2020, 3 marks

Marks	5	10	15	20	25
Number of students	3	7	f	9	6

SOLUTION

$\sum f = 25 + f$ and $\sum fx = 15 + 70 + 15f + 180 + 150 = 415 + 15f$

$\frac{415 + 15f}{25 + f} = 16$, so $415 + 15f = 400 + 16f$

Answer: $f = 15$

2 Mean of grouped data

4. The table gives the running times of some films. Find the mean running time by the step-deviation method. ICSE 2024, 3 marks

Duration (minutes)	100–110	110–120	120–130	130–140	140–150	150–160
Number of films	5	10	17	8	6	4

SOLUTION

Class marks: 105, 115, 125, 135, 145, 155. Take $A = 125$ and $h = 10$, so $u = \frac{x - 125}{10}$ is $-2, -1, 0, 1, 2, 3$.

$\sum f = 50$ and $\sum fu = -10 - 10 + 0 + 8 + 12 + 12 = 12$

Mean $= A + \frac{\sum fu}{\sum f} \times h = 125 + \frac{12}{50} \times 10$

Answer: 127.4 minutes

5. Find the mean of the marks below by the short-cut method. ICSE 2023, 3 marks

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	3	8	14	9	4	2

SOLUTION

Class marks: 5, 15, 25, 35, 45, 55. Take the assumed mean $A = 25$, so $d = x - 25$ is $-20, -10, 0, 10, 20, 30$.

$\sum f = 40$ and $\sum fd = -60 - 80 + 0 + 90 + 80 + 60 = 90$

Mean $= A + \frac{\sum fd}{\sum f} = 25 + \frac{90}{40}$

Answer: 27.25 marks

6. The table shows the number of patients attending a hospital on the days of a month. Find the mean by the short-cut method, taking the assumed mean as 45. Give the answer correct to 2 decimal places. ICSE 2019, 3 marks

Number of patients	10–20	20–30	30–40	40–50	50–60	60–70
Number of days	5	2	7	9	2	5

SOLUTION

Class marks: 15, 25, 35, 45, 55, 65. With $A = 45$, $d = x - 45$ is $-30, -20, -10, 0, 10, 20$.

$\sum f = 30$ and $\sum fd = -150 - 40 - 70 + 0 + 20 + 100 = -140$

Mean $= 45 + \dfrac{-140}{30} = 45 - 4.67$

Answer: 40.33 patients

7. The mean of the data below is 24. Find a. ICSE 2018, 3 marks

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	7	a	8	10	5

SOLUTION

Class marks: 5, 15, 25, 35, 45. $\sum f = 30 + a$ and $\sum fx = 35 + 15a + 200 + 350 + 225 = 810 + 15a$

$\dfrac{810 + 15a}{30 + a} = 24$, so $810 + 15a = 720 + 24a$

Answer: $a = 10$

8. Find the mean of the data below by the step-deviation method. ICSE 2017, 4 marks

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	10	9	25	30	16	10

SOLUTION

Class marks: 5, 15, 25, 35, 45, 55. Take $A = 35$ and $h = 10$, so u is $-3, -2, -1, 0, 1, 2$.

$\sum f = 100$ and $\sum fu = -30 - 18 - 25 + 0 + 16 + 20 = -37$

Mean $= 35 + \dfrac{-37}{100} \times 10 = 35 - 3.7$

Answer: 31.3 marks

3 Median of raw data

9. The numbers $27, 31, 46, 52, x, x+4, 71, 79, 85, 90$ are in ascending order and their median is 64. Find x.
(a) 60 (b) 61 (c) 62 (d) 66 ICSE 2024, 1 mark

SOLUTION

There are 10 numbers, so the median is the mean of the 5th and 6th: $\frac{x+(x+4)}{2} = x+2$.

$x+2 = 64$, so $x = 62$.

Answer: (c)

10. The marks (out of 10) of 40 students in a class test are given below. Find (a) the median (b) the mode.
ICSE 2019, 4 marks

Marks	1	2	3	4	5	6	7	8	9	10
Number of students	1	2	3	3	6	10	5	4	3	3

SOLUTION

Cumulative frequencies: 1, 3, 6, 9, 15, 25, 30, 34, 37, 40.

(a) $n = 40$ is even, so the median is the mean of the 20th and 21st values. Values 16 to 25 are all 6, so the median is 6.

(b) The highest frequency is 10, for the mark 6.

Answer: (a) median $= 6$ (b) mode $= 6$

11. The marks of 10 students, in ascending order, are $13, 35, 43, 46, x, x+4, 55, 61, 71, 80$. The median is 48. Find x, and then the mode. ICSE 2017, 3 marks

SOLUTION

With 10 values, the median is the mean of the 5th and 6th: $\frac{x+x+4}{2} = 48$, so $x = 46$.

The data is 13, 35, 43, 46, 46, 50, 55, 61, 71, 80. The value 46 occurs twice.

Answer: $x = 46$; mode $= 46$

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